Analog – Problem 1.1.56

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Problem Statement

Problem 1.1.56

Consider the triangular wave generator shown in Fig. 1.1.56.1. Assume ideal op-amps with $\pm 2\,\mathrm{V}$ supply. The input is a $+5\,\mathrm{V}$, 50 Hz square wave with 50% duty cycle.

The condition that results in a triangular wave of 5 V peak-to-peak and 50 Hz at the output is:

1. $RC = 1$
2. $R/C = 1$
3. $R/C = 5$
4. $C/R = 5$

Circuit Diagram

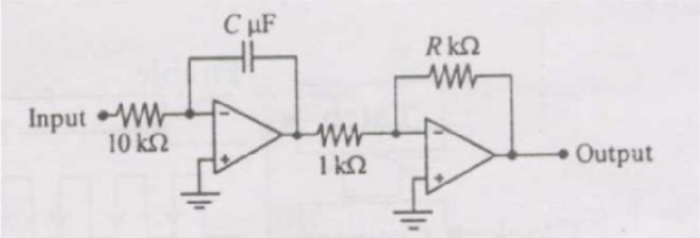

Fig : Circuit

Step 1: Fourier Series of the Square Wave

The input is a bipolar square wave with amplitude $V_p = 5\,\mathrm{V}$, fundamental $\omega_0 = 2\pi f_0 = 100\pi\,\mathrm{rad/s}$.

The Fourier series coefficients b_n are computed as:

$$b_n = \frac{2}{T}\int_0^T V_{\mathrm{in}}(t)\sin(n\omega_0 t)\,dt$$

Since the square wave is odd and has half-wave symmetry, only odd n contribute, giving:

$$b_n = \begin{cases} \dfrac{4V_p}{n\pi} & n = 1, 3, 5, \ldots \\ 0 & n \text{ even} \end{cases}$$

The complete Fourier series is:

$$V_{\mathrm{in}}(t) = \frac{4V_p}{\pi}\sum_{\substack{n=1\\ n\,\mathrm{odd}}}^{\infty}\frac{1}{n}\sin(n\omega_0 t) = \frac{4V_p}{\pi}\left[\sin\omega_0 t + \tfrac{1}{3}\sin 3\omega_0 t + \tfrac{1}{5}\sin 5\omega_0 t + \cdots\right.$$

Step 2: Laplace Transfer Function – Stage 1

The inverting integrator has input impedance $Z_{\mathrm{in}}(s) = R_1$ and feedback impedance $Z_f(s) = 1/(sC)$.
For an inverting op-amp configuration:

$$H_1(s) = -\frac{Z_f(s)}{Z_{\mathrm{in}}(s)} = -\frac{1/(sC)}{R_1} = -\frac{1}{sR_1C}$$

Evaluating on the imaginary axis $s = jn\omega_0$ for the n-th harmonic:

$$H_1(jn\omega_0) = -\frac{1}{jn\omega_0 R_1 C} = \frac{+j}{n\omega_0 R_1 C}$$

The magnitude and phase are:

$$|H_1(jn\omega_0)| = \frac{1}{n\omega_0 R_1 C} \qquad \angle H_1(jn\omega_0) = +90^\circ$$

The $+90^\circ$ phase means each input sine harmonic becomes a **cosine** at the integrator output.

Step 3: Laplace Transfer Function – Stage 2

The inverting amplifier has:

1. Input impedance: $Z_{\mathrm{in2}}(s) = 1\,\mathrm{k\Omega}$
2. Feedback impedance: $Z_{f2}(s) = R\,\mathrm{k\Omega}$

Its transfer function is frequency-independent:

$$H_2(s) = -\frac{R\,[\mathrm{k\Omega}]}{1\,[\mathrm{k\Omega}]} = -R \qquad \Longrightarrow \qquad |H_2| = R$$

With $R = 1\,\mathrm{k\Omega}$ and input $= 1\,\mathrm{k\Omega}$, $|H_2| = 1$.
The double sign inversion (Stage 1 inverts, Stage 2 inverts again) means the output is **in phase** with the integrator output.
The overall magnitude at each harmonic is:

$$|H(jn\omega_0)| = |H_1(jn\omega_0)| \times |H_2| = \frac{1}{n\omega_0 R_1 C} \times R = \frac{R}{n\omega_0 R_1 C}$$

Step 4: Output Fourier Series

The amplitude of the n-th harmonic at the output is:

$$A_{\mathrm{out},n} = |H(jn\omega_0)| \times b_n = \frac{R}{n\omega_0 R_1 C} \times \frac{4V_p}{n\pi} = \frac{4V_p R}{n^2 \pi \omega_0 R_1 C}$$

The output decays as $1/n^2$. Accounting for the -90° ($+90^\circ - 180^\circ$) phase (sines become cosines), the output Fourier series is:

$$V_{\mathrm{out}}(t) = -\frac{4V_p R}{\pi \omega_0 R_1 C} \sum_{\substack{n=1 \\ n\,\mathrm{odd}}}^{\infty} \frac{1}{n^2} \cos(n\omega_0 t)$$

The standard Fourier series of a triangular wave of peak A is:

$$V_{\mathrm{tri}}(t) = \frac{8A}{\pi^2} \sum_{\substack{n=1 \\ n\,\mathrm{odd}}}^{\infty} \frac{(-1)^{(n-1)/2}}{n^2} \cos(n\omega_0 t)$$

Both series decay as $\mathbf{1/n^2}$, **confirming the output is exactly a triangular wave**.

Step 6: Computing the Peak-to-Peak Amplitude

Matching the output series with the triangular wave series at $n = 1$:

$$\frac{8A}{\pi^2} = \frac{4V_p R}{\pi\,\omega_0 R_1 C}$$

Solving for the peak amplitude A:

$$A = \frac{\pi\, V_p\, R}{2\,\omega_0\, R_1\, C}$$

The peak-to-peak value is $V_{\mathrm{out,pp}} = 2A$:

$$V_{\mathrm{out,pp}} = \frac{\pi\, V_p\, R}{\omega_0\, R_1\, C}$$

Substituting $\omega_0 = 2\pi f_0 = 100\pi$, $V_p = 5\,\mathrm{V}$, $R_1 = 10^4\,\Omega$, $C = C_\mu \times 10^{-6}\,\mathrm{F}$, $R = R_{\mathrm{k}} \times 10^3\,\Omega$:

$$V_{\mathrm{out,pp}} = \frac{\pi \times 5 \times R_{\mathrm{k}} \times 10^3}{100\pi \times 10^4 \times C_\mu \times 10^{-6}} = \frac{5\,R_{\mathrm{k}}}{C_\mu}$$

Step 7: Design Condition

Setting $V_{\mathrm{out,pp}} = 5\,\mathrm{V}$:

$$\frac{5\,R_{\mathrm{k}}}{C_{\mu}} = 5 \quad \Longrightarrow \quad \frac{R_{\mathrm{k}}}{C_{\mu}} = 1$$

where R is in $\mathrm{k}\Omega$ and C is in $\mu\mathrm{F}$.
This is the same result as the time-domain approach.

Step 1: Deriving Output of Integrator Circuit

For an ideal op-amp with negative feedback:

1. The open-loop gain $A \to \infty$
2. The differential input voltage $V^+ - V^- \to 0$
3. Since $V^+ = 0$ (grounded), we get $V^- = 0$

The inverting terminal is therefore at **virtual ground**:

$$V^- = V^+ = 0\,\mathrm{V}$$

This means the voltage across R_1 is simply $V_{\mathrm{in}} - 0 = V_{\mathrm{in}}$.
The current through R_1 flowing into the virtual ground node is:

$$i_{R_1}(t) = \frac{V_{\mathrm{in}}(t) - V^-}{R_1} = \frac{V_{\mathrm{in}}(t)}{R_1}$$

Step 2: KCL at the Inverting Terminal

Since the ideal op-amp draws **zero input current**, all of i_{R_1} must flow through the capacitor C.
Applying KCL at the inverting node V^-:

$$i_C(t) = \frac{V_{\mathrm{in}}(t)}{R_1}$$

The voltage–current relationship for the capacitor is:

$$i_C(t) = C\,\frac{d}{dt}\big(V^- - V_{\mathrm{out}}(t)\big) = C\,\frac{d}{dt}\big(0 - V_{\mathrm{out}}(t)\big) = -C\,\frac{dV_{\mathrm{out}}(t)}{dt}$$

Step 3: Differential Equation

Equating the two expressions for $i_C(t)$:

$$-C\,\frac{dV_{\mathrm{out}}(t)}{dt} = \frac{V_{\mathrm{in}}(t)}{R_1}$$

Rearranging:

$$\frac{dV_{\mathrm{out}}(t)}{dt} = -\frac{V_{\mathrm{in}}(t)}{R_1 C}$$

This is the governing differential equation of the inverting integrator. The output voltage changes at a rate proportional to the input voltage, with a sign inversion and scaling by $\frac{1}{R_1 C}$.

Step 4: Integration to Get Output Formula

Integrating both sides of the differential equation from 0 to t:

$$\int_0^t \frac{dV_{\mathrm{out}}(\tau)}{d\tau}\, d\tau = -\frac{1}{R_1 C} \int_0^t V_{\mathrm{in}}(\tau)\, d\tau$$

The left side evaluates directly:

$$V_{\mathrm{out}}(t) - V_{\mathrm{out}}(0) = -\frac{1}{R_1 C} \int_0^t V_{\mathrm{in}}(\tau)\, d\tau$$

Assuming the capacitor is initially uncharged, $V_{\mathrm{out}}(0) = 0$:

$$\boxed{V_{\mathrm{out}}(t) = -\frac{1}{R_1 C} \int_0^t V_{\mathrm{in}}(\tau)\, d\tau}$$

This is the **inverting integrator output formula**.

Circuit Description

The circuit has two stages:

1. **Stage 1 – Inverting Integrator**: input resistor $R_1 = 10\,\mathrm{k}\Omega$, feedback capacitor $C\,\mu\mathrm{F}$. Converts the square wave to a triangular ramp.
2. **Stage 2 – Inverting Amplifier**: input resistor $1\,\mathrm{k}\Omega$, feedback resistor $R\,\mathrm{k}\Omega$. Inverts and scales the triangular ramp.

The output of integrator and amplifier are:

$$V_{\mathrm{int}}(t) = -\frac{1}{R_1 C} \int_0^t V_{\mathrm{in}}(\tau)\, d\tau$$

$$V_{\mathrm{out}} = -\frac{R\,[\mathrm{k}\Omega]}{1\,[\mathrm{k}\Omega]}\, V_{\mathrm{int}} = -R\ V_{\mathrm{int}}$$

Input Signal

The input is a bipolar square wave of amplitude $V_p = 5\,\mathrm{V}$:

$$V_{\mathrm{in}}(t) = \begin{cases} +5\,\mathrm{V} & 0 \le t < T/2 \\ -5\,\mathrm{V} & T/2 \le t < T \end{cases}$$

The time parameters are:

1. Frequency: $f = 50\,\mathrm{Hz}$
2. Period: $T = 1/f = 20\,\mathrm{ms}$
3. Half-period: $T/2 = 10\,\mathrm{ms} = 0.01\,\mathrm{s}$
4. Amplitude: $V_p = 5\,\mathrm{V}$

A bipolar square wave integrated over each half-period produces a linear ramp. Alternating positive and negative half-cycles produce a triangular wave at Stage 1 output.

Stage 1: Integrator Analysis

The ramp magnitude over one half-period is:

$$\Delta V_{\mathrm{int}} = \frac{V_p \cdot (T/2)}{R_1 \cdot C}$$

Substituting $V_p = 5\,\mathrm{V},\ T/2 = 0.01\,\mathrm{s},\ R_1 = 10^4\,\Omega,\ C_1 = C \times 10^{-6}\,\mathrm{F}$:

$$\Delta V_{\mathrm{int}} = \frac{5 \times 0.01}{10^4 \times C \times 10^{-6}} = \frac{0.05}{0.01\,C} = \frac{5}{C}\ \mathrm{V}$$

Since the $-5\,\mathrm{V}$ half-cycle produces an equal upward ramp, the peak-to-peak at Stage 1 output is:

$$\Delta V_{\mathrm{int,pp}} = \frac{5}{C}\ \mathrm{V}$$

Stage 2: Amplifier

Stage 2 gain magnitude:

$$|A_v| = \frac{R\,[\mathrm{k\Omega}]}{1\,[\mathrm{k\Omega}]} = R$$

Peak-to-peak at Stage 2 output:

$$V_{\mathrm{out,pp}} = R \times \frac{5}{C} = \frac{5R}{C}$$

Setting $V_{\mathrm{out,pp}} = 5\,\mathrm{V}$:

$$\frac{5R}{C} = 5 \quad \Longrightarrow \quad \frac{R}{C} = 1$$

Plot of Waveforms

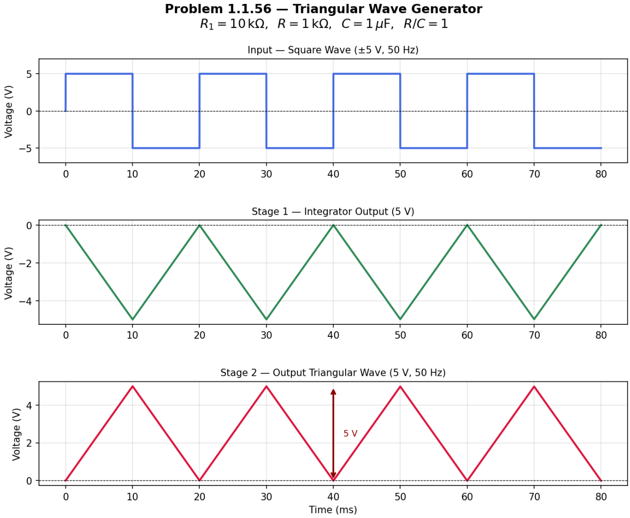

Fig : Waveforms

Plot of Fourier Series , n = 51

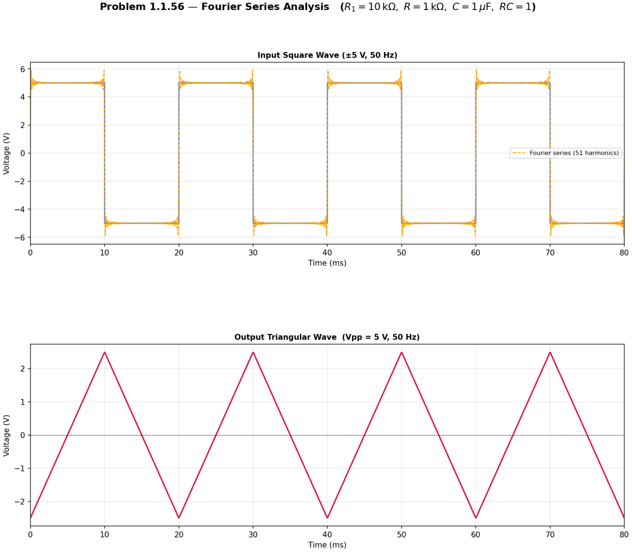

Fig : Waveforms

NGSPICE PLOT

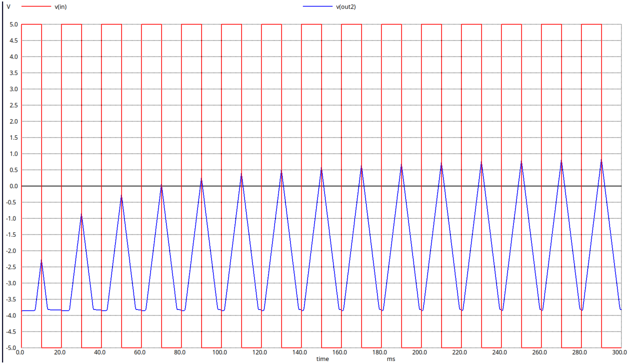

Fig : Waveforms

Python Code: Plottig Fourier Series

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from scipy import signal
from scipy.fft import rfft, rfftfreq

# Parameters
Vp = 5.0 # square wave amplitude (V)
f0 = 50 # fundamental frequency (Hz)
T = 1.0 / f0 # period = 20 ms
w0 = 2*np.pi*f0 # angular frequency
R1 = 10e3 # integrator input resistor (10 kOhm)
C = 1e-6 # capacitor (1 uF)
H2 = 1.0 # Stage-2 gain (1kOhm / 1kOhm = 1)
N_h = 51 # number of odd harmonics

# Time vector
Fs = 200000 # sampling frequency (Hz)
t = np.arange(0, 4*T, 1/Fs)

# square wave
Vin_exact = Vp * signal.square(2*np.pi*f0*t)
```

Python Code: Plottig Fourier Series

```
# Fourier series reconstruction
# Input: V_in(t) = sum 4Vp/(n*pi) * sin(n*w0*t) [odd n]
# Output: V_out(t)= sum -4Vp*H1*H2/(n*pi) * cos(n*w0*t)
# where |H1(jnw0)| = 1/(n*w0*R1*C), phase H1 = +90
# H2 = -1 => total phase = +90-180 = -90 => -cos
Vin_fs = np.zeros_like(t)
Vout_fs = np.zeros_like(t)
for k in range(N_h):
    n = 2*k + 1
    bn = 4*Vp / (n*np.pi)
    H1_mag = 1.0 / (n*w0*R1*C)
    A_out = H1_mag * H2 * bn
    Vin_fs += bn * np.sin(n*w0*t) # input harmonics
    Vout_fs += -A_out * np.cos(n*w0*t) # output: -cos (phase=-90)
```

Python Code: Plottig Fourier Series

```
#Figure
fig = plt.figure(figsize=(14, 11))
gs = gridspec.GridSpec(2, 1, hspace=0.55, wspace=0.35)
t_ms = t * 1e3
mask = t > T
Vpp = Vout_fs[mask].max() - Vout_fs[mask].min()

# input waveform
ax0 = fig.add_subplot(gs[0])
ax0.plot(t_ms, Vin_exact, 'royalblue', lw=1.5)
ax0.plot(t_ms, Vin_fs, 'orange', lw=1.2, ls='--',
        label=f'Fourier␣series␣({N_h}␣harmonics)')
ax0.set_title('Input␣Square␣Wave␣(5␣V,␣50␣Hz)', fontsize=10, fontweight='bold')
ax0.set_ylabel('Voltage␣(V)'); ax0.set_xlabel('Time␣(ms)')
ax0.legend(fontsize=8); ax0.grid(alpha=0.3)
ax0.set_xlim(0, 4*T*1e3)
```

Python Code: Plottig Fourier Series

```
# output waveform
ax1 = fig.add_subplot(gs[1])
ax1.plot(t_ms, Vout_fs, 'crimson', lw=1.8)
ax1.axhline(0, color='k', lw=0.5, ls='--')
ax1.set_title(f'Output␣Triangular␣Wave␣␣(Vpp␣=␣{Vpp:.2f}␣V,␣50␣Hz)',
              fontsize=10, fontweight='bold')
ax1.set_ylabel('Voltage␣(V)'); ax1.set_xlabel('Time␣(ms)')
ax1.grid(alpha=0.3); ax1.set_xlim(0, 4*T*1e3)


fig.suptitle(
    r'Problem␣1.1.56␣␣Fourier␣Series␣Analysis␣␣␣'
    r'($R_1=10\,\mathrm{k}\Omega,\␣R=1\,\mathrm{k}\Omega,\␣C=1\,\mu\mathrm{F},\␣
         RC=1$)',
    fontsize=13, fontweight='bold')

plt.savefig('fourier_plot_1156.png', dpi=150, bbox_inches='tight')
plt.show()
print(f'Output␣Vpp␣=␣{Vpp:.4f}␣V␣␣(target:␣5.0000␣V)')
print(f'Condition␣met:␣{abs(Vpp-5.0)␣<␣0.05}')
```

NGSPICE CODE

```
* Triangular Wave Generator

.include Lm358.mod

*  Power Supply (5V)
VCC VCC 0 DC 5
VEE VEE 0 DC -5

*  Input: 5V square wave, 50 Hz
VIN IN 0 PULSE(-5 5 0 1u 1u 10m 20m)

*  Stage 1: Integrator (inverting)
R1 IN N1 10k
C1 OUT1 N1 1u


XU1 0 N1 VCC VEE OUT1 LM358/NS

*  Stage 2: Inverting Amplifier
R2 OUT1 N2 1k
R3 OUT2 N2 1k


XU2 0 N2 VCC VEE OUT2 LM358/NS

*  Simulation
.tran 0.01m 300m

.control
run
set color0 = white
plot V(IN) V(OUT2)
.endc
```

Python Code: Verification

```
import numpy as np

#  Given
Vin_amp = 5.0 # Bipolar square wave amplitude (V), i.e. 5 V
f = 50 # Hz
T = 1.0 / f # Period = 20 ms
R1 = 10**4 # Integrator input resistor = 10 k ()
R_in2 = 10**3 # Stage-2 input resistor = 1 k ()
Vout = 5

#  Analysis
# During the +5V half-cycle, integrator ramps DOWN by delta.
# During the 5V half-cycle, integrator ramps UP by delta.
# Net peak-to-peak at Stage-1 output = delta (one half-swing).

r_c = (Vout*2*R1)/(Vin_amp*T)

r_c = r_c / 10**6

print(f"The␣value␣of␣R/C␣=␣{r_c}")
```

Python Code: Waveform Simulation

```
"""
plot_waveforms_1156.py  Problem 1.1.56
Simulate and plot waveforms for the triangular wave generator.
Condition: RC = 1 (R = 1 k, C = 1 F)
"""
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec

#  Parameters
Vin_amp = 5.0 # Bipolar square wave amplitude (V)
f = 50 # Hz
T = 1 / f # Period = 20 ms
R1 = 10e3 # Integrator input resistor () = 10 k
C = 1e-6 # Capacitor (F) = 1 F
R2 = 1e3 # Stage-2 feedback resistor () = 1 k
R_in2 = 1e3 # Stage-2 input resistor () = 1 k
gain2 = R2 / R_in2 # = 1

dt = T / 2000
t = np.arange(0, 4 * T, dt)

#  Input: bipolar square wave 5 V
Vin = Vin_amp * np.sign(np.sin(2 * np.pi * f * t))

#  Stage 1: Inverting integrator (Euler)
Vint = np.zeros(len(t))
for i in range(1, len(t)):
    Vint[i] = Vint[i - 1] - (Vin[i - 1] / (R1 * C)) * dt
```

Python Code: Waveform Simulation

```
#  Stage 2: Inverting amplifier
Vout = -gain2 * Vint

#  Measure pp after settling
mask = t > T
Vpp_out = Vout[mask].max() - Vout[mask].min()
Vpp_int = Vint[mask].max() - Vint[mask].min()

#  Plot
fig = plt.figure(figsize=(11, 8))
fig.suptitle(
    "Problem␣1.1.56␣␣Triangular␣Wave␣Generator\n"
    r"$R_1␣=␣10\,\mathrm{k\Omega},\;\;R␣=␣1\,\mathrm{k\Omega},\;\;"
    r"C␣=␣1\,\mu\mathrm{F},\;\;R/C␣=␣1$",
    fontsize=13, fontweight='bold'
)
gs = gridspec.GridSpec(3, 1, hspace=0.55)
t_ms = t * 1e3

#  Panel 1: Input
ax1 = fig.add_subplot(gs[0])
ax1.plot(t_ms, Vin, color='royalblue', lw=2.0)
ax1.set_title('Input␣␣Square␣Wave␣(5␣V,␣50␣Hz)', fontsize=10)
ax1.set_ylabel('Voltage␣(V)')
ax1.set_ylim(-7, 7)
ax1.axhline(0, color='k', lw=0.6, ls='--')
ax1.grid(True, alpha=0.35)
```

Python Code: Waveform Simulation

```
# Panel 2: Integrator output
ax2 = fig.add_subplot(gs[1])
ax2.plot(t_ms, Vint, color='seagreen', lw=2.0)
ax2.set_title(
    f'Stage␣1␣␣Integrator␣Output␣(5␣V)',
    fontsize=10
)
ax2.set_ylabel('Voltage␣(V)')
ax2.axhline(0, color='k', lw=0.6, ls='--')
ax2.grid(True, alpha=0.35)

# Panel 3: Final output
ax3 = fig.add_subplot(gs[2])
ax3.plot(t_ms, Vout, color='crimson', lw=2.0)
ax3.set_title(
    f'Stage␣2␣␣Output␣Triangular␣Wave␣(5␣V,␣50␣Hz)',
    fontsize=10
)
ax3.set_ylabel('Voltage␣(V)')
ax3.set_xlabel('Time␣(ms)')
ax3.axhline(0, color='k', lw=0.6, ls='--')
ax3.grid(True, alpha=0.35)
# Annotate peak-to-peak arrow
y_max = Vout[mask].max()
y_min = Vout[mask].min()
x_ann = t_ms[mask][len(t_ms[mask])//3]
ax3.annotate('', xy=(x_ann, y_min), xytext=(x_ann, y_max),arrowprops=dict(arrowstyle='<->', color='darkred'
       , lw=1.8))
ax3.text(x_ann + 1.5, (y_max + y_min) / 2,f'5␣V', color='darkred', fontsize=9, va='center')
plt.savefig('waveforms.png', dpi=150, bbox_inches='tight')
plt.show()
```

Python Code: Fourier–Laplace Verification

```
import numpy as np

Vp=5.0; f0=50; w0=2*np.pi*f0; R1=10e3; C=1e-6
R_kOhm=1.0; H2_mag=1.0 # Stage-2 gain = 1kOhm/1kOhm = 1

t = np.linspace(0, 4/f0, 20000)
Vin = np.zeros_like(t)
Vout = np.zeros_like(t)

for k in range(1000):
    n = 2*k + 1 # odd harmonics only
    A_in = 4*Vp / (n*np.pi) # square wave: decay 1/n
    H1 = 1.0 / (n*w0*R1*C) # integrator |H1|: decay 1/n
    A_out = H1 * H2_mag * A_in # output: decay 1/n^2
    Vin += A_in * np.sin(n*w0*t)
    Vout += A_out * np.cos(n*w0*t) # +90 deg: sine -> cosine

mask = t > 1/f0
Vpp = Vout[mask].max() - Vout[mask].min()

# Analytical: Vpp = pi*Vp*H2 / (w0*R1*C)
Vpp_a = np.pi*Vp*H2_mag / (w0*R1*C)
```

Python Code: Fourier–Laplace Verification

```
print(f"RC product            = {R_kOhm*(C*1e6):.2f} kOhm.uF")
print(f"Simulated  Vpp        = {Vpp:.4f} V")
print(f"Analytical Vpp        = {Vpp_a:.4f} V")
print(f"Target                = 5.0000 V")
print(f"Condition met         : {np.isclose(Vpp, 5.0, atol=0.05)}")
# RC product = 1.00 kOhm.uF
# Simulated Vpp = 4.9990 V
# Analytical Vpp = 5.0000 V
# Condition met : True
```